

Sketching Tangent Curves

Use the applet to help you understand the shape, then complete the sketches by hand.

Key form:

$$y = a \tan(b(x - c)) + d$$

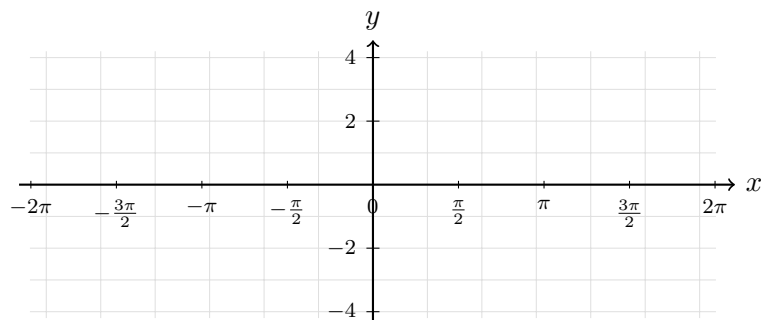
Feature	How to find it
Period	$\frac{\pi}{ b }$
Principal line	$y = d$
Centre point	(c, d)
Vertical asymptotes	$x = c + \frac{\frac{\pi}{2} + k\pi}{b}$, where k is an integer
Effect of a	Vertical stretch or reflection. If $a < 0$, the curve is reflected.

Part A: Basic tangent curves

For each function, find the main features and sketch the graph on the axes.

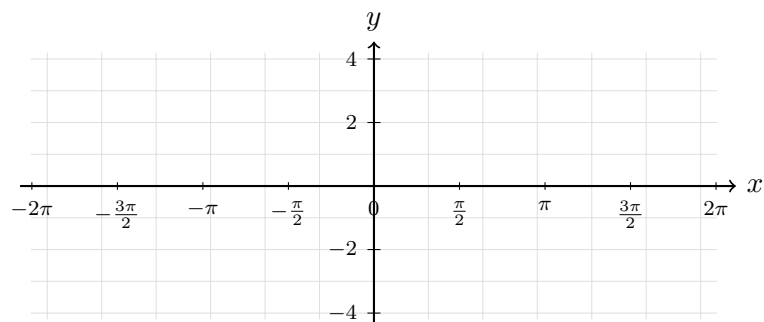
1. $y = \tan x$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



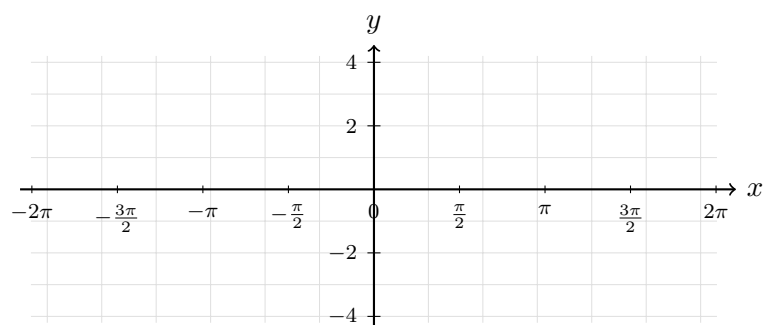
2. $y = 2 \tan x$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



3. $y = -\tan x$

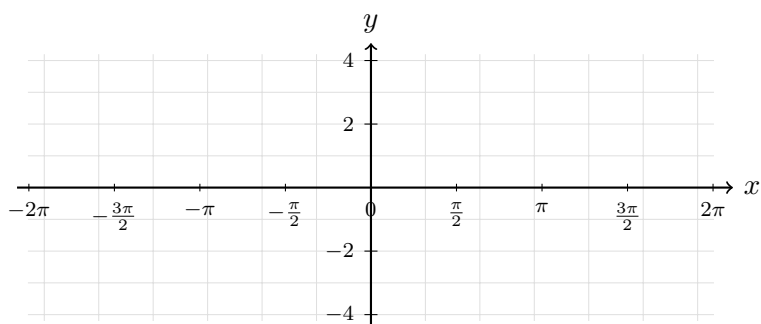
Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



Part B: Changing the period

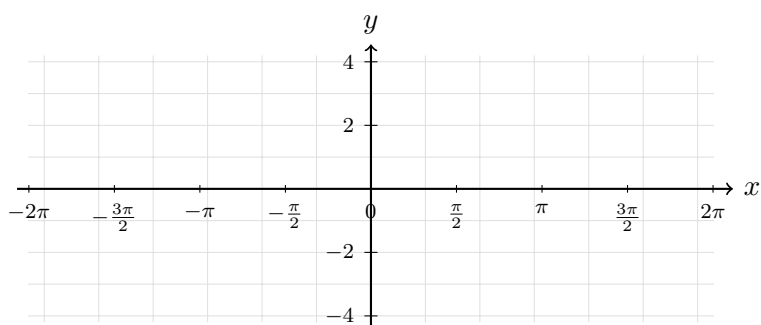
4. $y = \tan(2x)$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



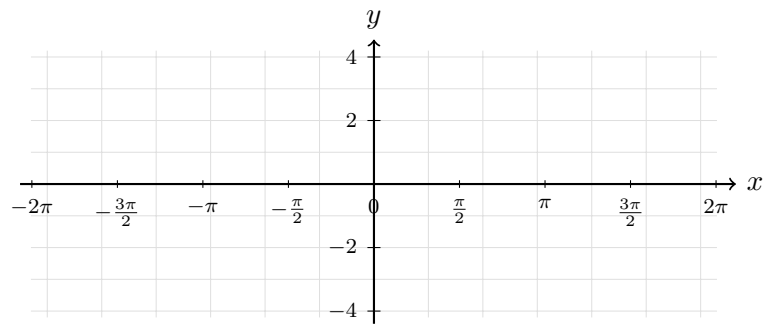
5. $y = \tan\left(\frac{1}{2}x\right)$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



6. $y = 3 \tan(2x)$

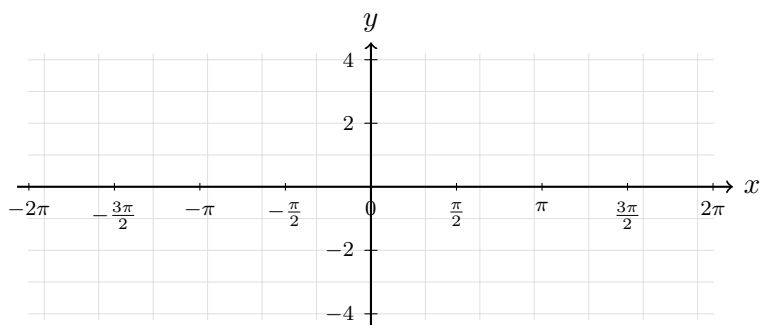
Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



Part C: Translations

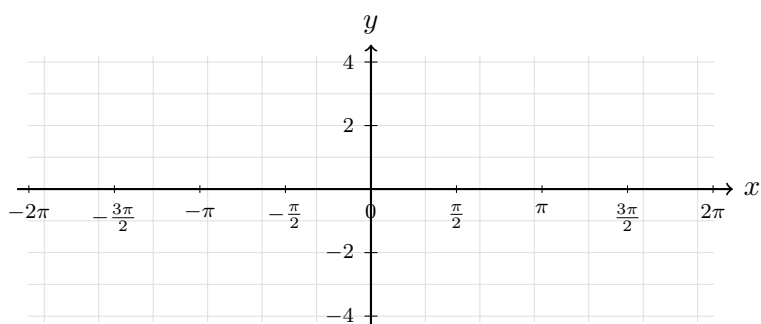
7. $y = \tan\left(x - \frac{\pi}{4}\right)$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



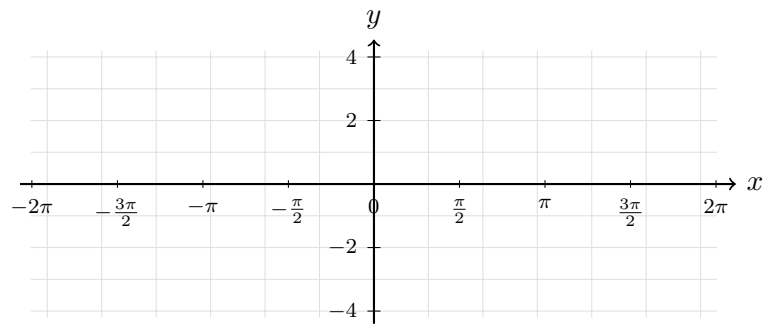
8. $y = \tan x + 2$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



9. $y = \tan\left(x + \frac{\pi}{2}\right) - 1$

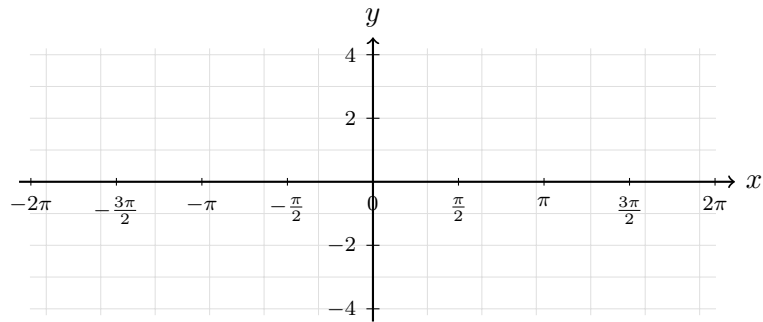
Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



Part D: Mixed transformations

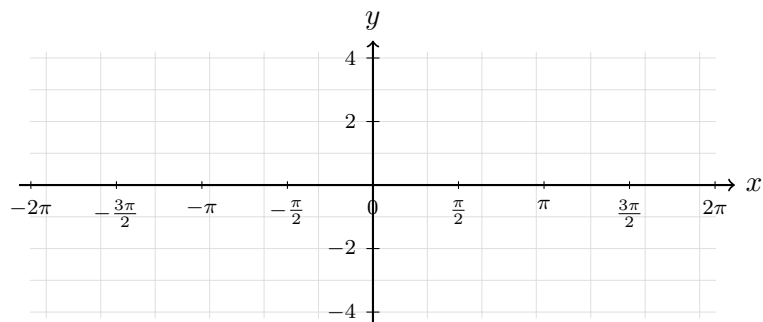
10. $y = 2 \tan \left(x - \frac{\pi}{3} \right) + 1$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



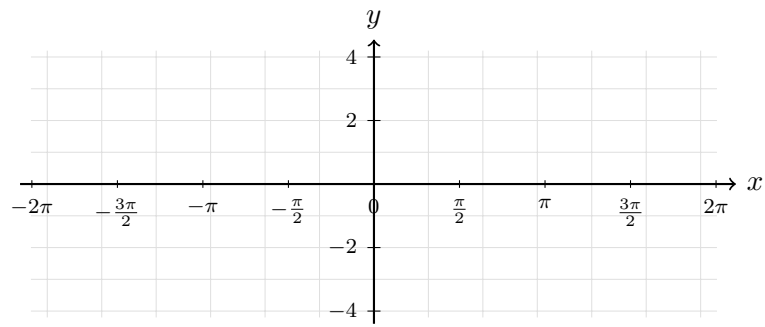
11. $y = -\tan \left(2\left(x + \frac{\pi}{4}\right) \right)$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



12. $y = \frac{1}{2} \tan \left(3\left(x - \frac{\pi}{6}\right) \right) - 2$

Period	
Principal line	
Centre point	
Vertical asymptotes	
Important points	



Part E: Identify the tangent curve

For each description, write a possible equation in the form

$$y = a \tan(b(x - c)) + d.$$

1. Period = π , principal line $y = 3$, centre point $(0, 3)$.

Equation: _____

2. Period = $\frac{\pi}{2}$, principal line $y = 0$, reflected in the x -axis.

Equation: _____

3. Period = 2π , principal line $y = -1$, centre point $(0, -1)$.

Equation: _____

4. Period = π , centre point $(\frac{\pi}{4}, 2)$, vertical stretch factor 3.

Equation: _____

5. Period = $\frac{\pi}{3}$, principal line $y = 1$, centre point $(-\frac{\pi}{6}, 1)$.

Equation: _____

Part F: Explain the link with sine and cosine

Use the identity

$$\tan x = \frac{\sin x}{\cos x}.$$

1. Why do the zeroes of $\sin x$ become zeroes of $\tan x$?

2. Why do the zeroes of $\cos x$ become vertical asymptotes of $\tan x$?

3. Explain why $y = \tan x$ repeats every π , not every 2π .

Teacher Answer Key

Sketches should show correct shape, principal line, centre point, and vertical asymptotes. Exact visual scale is less important than correct features.

1. $y = \tan x$: period π , principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \frac{\pi}{2} + k\pi$.
2. $y = 2 \tan x$: period π , principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \frac{\pi}{2} + k\pi$. Steeper than $y = \tan x$.
3. $y = -\tan x$: period π , principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \frac{\pi}{2} + k\pi$. Reflected in the x -axis.
4. $y = \tan(2x)$: period $\frac{\pi}{2}$, principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \frac{\pi}{4} + \frac{k\pi}{2}$.
5. $y = \tan\left(\frac{1}{2}x\right)$: period 2π , principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \pi + 2k\pi$.
6. $y = 3 \tan(2x)$: period $\frac{\pi}{2}$, principal line $y = 0$, centre $(0, 0)$, asymptotes $x = \frac{\pi}{4} + \frac{k\pi}{2}$. Vertically stretched.
7. $y = \tan\left(x - \frac{\pi}{4}\right)$: period π , principal line $y = 0$, centre $\left(\frac{\pi}{4}, 0\right)$, asymptotes $x = \frac{3\pi}{4} + k\pi$.
8. $y = \tan x + 2$: period π , principal line $y = 2$, centre $(0, 2)$, asymptotes $x = \frac{\pi}{2} + k\pi$.
9. $y = \tan\left(x + \frac{\pi}{2}\right) - 1$: period π , principal line $y = -1$, centre $\left(-\frac{\pi}{2}, -1\right)$, asymptotes $x = k\pi$.
10. $y = 2 \tan\left(x - \frac{\pi}{3}\right) + 1$: period π , principal line $y = 1$, centre $\left(\frac{\pi}{3}, 1\right)$, asymptotes $x = \frac{\pi}{3} + \frac{\pi}{2} + k\pi = \frac{5\pi}{6} + k\pi$.
11. $y = -\tan\left(2\left(x + \frac{\pi}{4}\right)\right)$: period $\frac{\pi}{2}$, principal line $y = 0$, centre $\left(-\frac{\pi}{4}, 0\right)$, asymptotes $x = -\frac{\pi}{4} + \frac{\pi}{4} + \frac{k\pi}{2} = \frac{k\pi}{2}$. Reflected.
12. $y = \frac{1}{2} \tan\left(3\left(x - \frac{\pi}{6}\right)\right) - 2$: period $\frac{\pi}{3}$, principal line $y = -2$, centre $\left(\frac{\pi}{6}, -2\right)$, asymptotes $x = \frac{\pi}{6} + \frac{\pi}{6} + \frac{k\pi}{3} = \frac{\pi}{3} + \frac{k\pi}{3}$.
13. Possible answer: $y = \tan x + 3$.
14. Possible answer: $y = -\tan(2x)$.
15. Possible answer: $y = \tan\left(\frac{1}{2}x\right) - 1$.
16. Possible answer: $y = 3 \tan\left(x - \frac{\pi}{4}\right) + 2$.
17. Since period $= \frac{\pi}{3}$, $b = 3$. Possible answer:

$$y = \tan\left(3\left(x + \frac{\pi}{6}\right)\right) + 1.$$

18. Since $\tan x = \frac{\sin x}{\cos x}$, when $\sin x = 0$ and $\cos x \neq 0$, the fraction equals 0, so $\tan x = 0$.
19. Since $\tan x = \frac{\sin x}{\cos x}$, when $\cos x = 0$, the denominator is zero. This creates a vertical asymptote.
20. $\tan x$ repeats every π because

$$\tan(x + \pi) = \frac{\sin(x + \pi)}{\cos(x + \pi)} = \frac{-\sin x}{-\cos x} = \frac{\sin x}{\cos x} = \tan x.$$